WEAVE: Overcoming the Identity Shortcut in Style Transfer via 3-Minute Wavelet-Decoupled Flow Matching

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Abstract

Current style transfer methods face two primary challenges. First, state-of-the-art diffusion models, along with training-free methods built on them, require more than 60 GB of VRAM and therefore cannot run on consumer GPUs. Second, lightweight methods without these generative backbones often underperform a trivial identity mapping, which scores high on traditional metrics like ArtFID and LPIPS but was rarely evaluated as a baseline.

We empirically observe that frequency dominance in shared representation spaces steers lightweight models toward identity-like solutions: low-frequency structural energy overwhelms high-frequency style gradients, a pattern we confirm through gradient-spectrum analysis and per-subband style-transfer measurements.

We argue that an effective style transfer method should outperform the identity baseline while remaining computationally affordable. To this end, we propose WEAVE: a single-level Haar wavelet transform splits the VAE latent into a low-frequency structure band and high-frequency style bands; a compact rectified-flow network learns a structure-preserving velocity field; and endpoint statistical alignment (AdaIN) injects the target style into the high-frequency subbands. This decoupling separates content transport from style injection, letting WEAVE train in 3 minutes on a single consumer GPU.

We also introduce an identity-floor-aware evaluation protocol, under which WEAVE clears the identity baseline with only 903K trainable parameters (plus a frozen 84M VAE), making high-quality style transfer both effective and affordable.

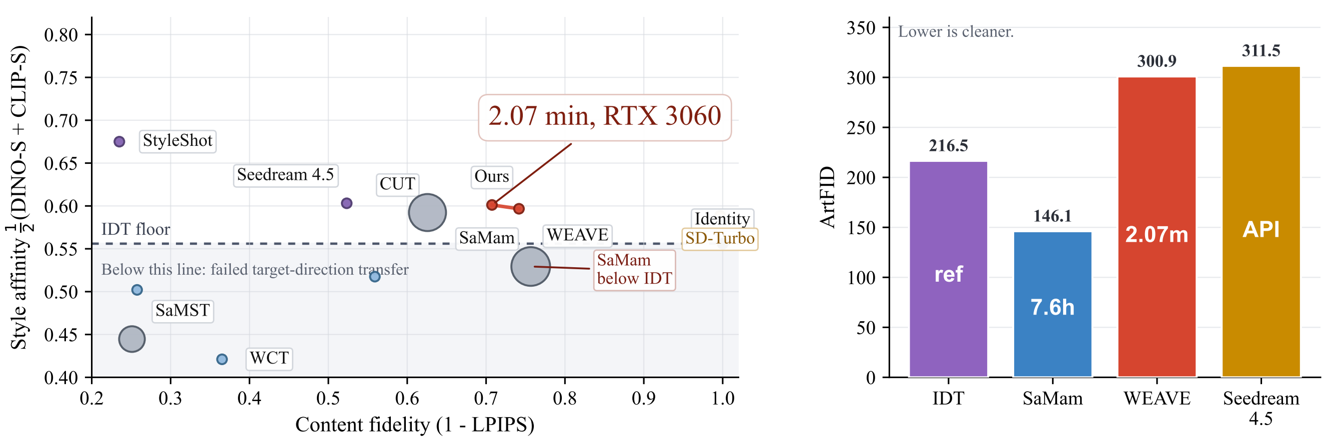

Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: style affinity $\frac{1}{2}(\text{DINO-S} + \text{CLIP-S})$ versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned). The dashed line marks the IDT floor (0.556); methods below it have not moved in the requested direction. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

Introduction

Style transfer faces two problems in deployment and evaluation. On one hand, state-of-the-art diffusion models (e.g., FLUX.2 and Qwen-image) and their training-free variants demand more than 60 GB of GPU memory, making them unusable on consumer hardware. On the other hand, certain lightweight alternatives that avoid these heavy generative backbones can struggle to achieve genuine style transfer. Specifically, the style affinity of outputs from certain lightweight models can fall below that of the unchanged source image (i.e., the identity mapping). This problem has been hidden by conventional evaluation protocols, such as LPIPS and ArtFID, which heavily reward structural reconstruction but rarely include the identity mapping as an explicit baseline. As visualized in Figure 1 (Left), when the identity mapping is explicitly plotted, modern lightweight methods like SaMam clearly fall below this natural baseline. To fix this, we introduce the Identical-Image Transfer (IDT) calibration, transforming standard metrics into floor-calibrated measures.

Why do recent lightweight architectures, despite their advanced designs, consistently fall into this identity trap? We argue that the root cause is not a lack of model capacity, but an inherent geometric conflict within shared representation spaces. In typical flow-matching frameworks, the model predicts a velocity field that transports the source latent toward the target. However, natural images follow a $1/f$ spectral distribution, meaning the low-frequency structural energy (e.g., global layouts and object shapes) is orders of magnitude stronger than the high-frequency stylistic textures (e.g., brushstrokes and material patterns). When minimizing reconstruction loss in shared representation spaces, the optimizer naturally prioritizes fitting the massive low-frequency errors. Consequently, the subtle style-conditioned gradients are overwhelmed, pushing the model toward a style-agnostic identity mapping. This is why simply scaling up parameters can fail: structure and style compete in the same coordinate basis.

To solve the conflict between structure and style, we propose WEAVE. We operate in the latent space of a pre-trained diffusion VAE, which compresses images into a clean, low-dimensional space, but discard the heavy diffusion network and train only a tiny (903K-parameter) velocity network. A single-level Haar wavelet transform separates the latent into structure (LL) and texture (LH/HL/HH); style is injected only through the high-frequency subband statistics (endpoint AdaIN), which stay anchored to their original spatial layout by construction. The style-injection operator never touches the LL band, so content structure is preserved throughout the 8-step integration. This streamlined design lets WEAVE clear the identity baseline and train in 3 minutes on a single consumer GPU; because the alignment operates directly on latent features, it also serves as a training-free plugin for large pre-trained models, which we validate on a frozen SD1.5 image-to-image pipeline (Section).

Contributions. (1) **Evaluation Protocol:** We introduce the Identical-Image Transfer (IDT) calibration, exposing the blind spot where lightweight models underperform the unchanged source. (2) **Representation and Methodology:** We empirically observe that frequency dominance in shared representation spaces causes low-frequency structures to overshadow high-frequency textures, driving models toward the identity shortcut, and propose WEAVE, a wavelet-decoupled framework that achieves genuine style transfer with only 903K trainable parameters. Mechanism experiments (gradient-spectrum analysis, per-subband style-transfer ratios, and inference-time module swapping) provide causal evidence for this functional specialization. (3) **Efficiency and Generalization:** WEAVE trains in just 3 minutes on a single consumer GPU, and its latent operations are a pluggable, training-free mechanism for large diffusion backbones, which we validate by applying them to a frozen SD1.5 image-to-image pipeline (Section).

Related Work

Evolution of Style Transfer

Early style transfer relied on optimizing pixel spaces (Gatys, Ecker, and Bethge 2015) or matching feature statistics in pre-trained discriminative networks, such as AdaIN (Huang and Belongie 2017) and WCT (Li et al. 2017). These methods are fast but produce artifacts when content and style share similar low-frequency statistics. Subsequent works introduced GANs and contrastive learning for unpaired translation, notably CUT (Park et al. 2020), which learns a content-to-style mapping without paired data but still requires tens of minutes of training per domain pair. Diffusion models moved style transfer from explicit feature transformation to generative editing in latent spaces (Rombach et al. 2022). Recently, the field has been dominated by heavy generative priors based on latent diffusion models. State-of-the-art backbones like FLUX.2 and Seedream, along with training-free attention manipulation techniques like StyleID (Chung, Hyun, and Heo 2024) and StyleAligned (Hertz et al. 2024), achieve strong style transfer performance by leveraging massive pre-trained representations. However, these approaches require 60 GB+ VRAM, making them inaccessible on consumer hardware and motivating the lightweight direction we pursue.

Lightweight Architectures and Evaluation Protocols

To reduce computational costs, modern lightweight architectures, such as Mamba-based SaMST and SaMam (Liu et al. 2024, 2025), have been explored. These methods train small networks on pixel or feature spaces but, as we show, can fall below the identity baseline on direction-aware metrics. While these methods improve efficiency, standard evaluation protocols (Zhang et al. 2018; Wright and Ommer 2022) heavily reward source reconstruction and historically omit the identity mapping as an explicit baseline. This evaluation blind spot allows methods that merely preserve content to appear competitive. Our IDT calibration addresses this by explicitly including the identity mapping, transforming CLIP-S and LPIPS into direction-aware metrics that expose when a method has not moved toward the target style.

Spectral Bias and Frequency-aware Optimization

Neural networks exhibit a well-documented spectral bias (Rahaman et al. 2019), preferentially fitting low-frequency signals. Combined with the $1/f$ spectral distribution of natural images, this observation motivates frequency-aware optimization strategies. Prior work has explored spectral decomposition for image restoration and generation, but the connection between spectral bias and the identity shortcut in style transfer has not been explicitly drawn. Our work shows that this bias is not merely a training nuisance but a structural cause of the identity trap: when content and style share the same coordinate basis, the optimizer's natural preference for low-frequency fidelity actively suppresses style transfer.

Wavelet and Latent-space Alignment

Wavelet transforms have been widely adopted in image restoration and generative modeling to separate multi-scale information, and have been adapted for style transfer (Yoo

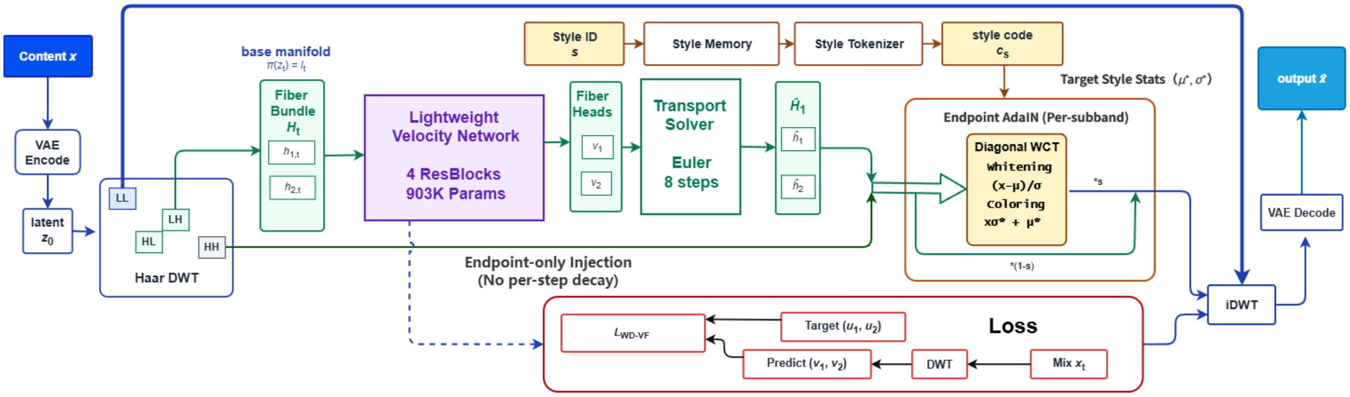

Figure 2: **WEAVE architecture**. A single-level Haar transform exposes the LL/LH/HL/HH subbands of the SD1.5 VAE latent. The compact velocity network v_θ ($\sim$903K) predicts motion for LL/LH/HL via three heads, while HH carries no head and is architecturally frozen. Training combines band-weighted rectified-flow matching (8-step Euler) with endpoint high-frequency statistical alignment (per-subband AdaIN), applied once at the final step on LH/HL/HH with scales 0.3/0.3/0.5 and LL excluded. Style conditioning enters only the shared backbone (weak, λ_{LL}=0.3); a style-memory cross-attention branch is auxiliary.

et al. 2019) and diffusion models (Phung, Dao, and Tran 2023). Concurrently, latent diffusion models (Rombach et al. 2022) introduced perceptual latent spaces that compress high-dimensional pixels into regularized representations. This regularized distribution has facilitated statistical alignment methods such as whitening-and-coloring transforms (Li et al. 2017), making distribution matching more stable in latent spaces. Our contribution is to combine these two threads—wavelet decomposition and latent-space statistical alignment—within a flow-matching framework, using the wavelet basis to structurally separate content transport from style injection rather than treating them as competing objectives in a shared space.

Method

We describe WEAVE following the logic of the Introduction: first, where to learn the transfer (VAE latent space); second, how to separate conflicting signals (wavelet decomposition); third, how to route style information while protecting content (frequency-aware architecture and endpoint AdaIN); and fourth, how the learned transport and the statistical aligner cooperate at inference.

Framework Setup

Let $x_c \in \mathbb{R}^{H\times W\times 3}$ be a content image and $s \in \{1, \dots, K\}$ a target style domain. We operate in the latent space of the Stable Diffusion v1.5 EMA VAE (Rombach et al. 2022). We write $z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4\times 32\times 32}$. During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment

$$z_t = (1-t)z_0 + tz_1, \quad u_t = z_1 - z_0, \quad t \sim \mathcal{U}([0,1]). \tag{1}$$

We choose rectified flow over DDPM-style diffusion because its straight trajectory is well-approximated by a small number of Euler steps: 8 steps suffice for convergence (Section), avoiding the thousands of denoising iterations typical of diffusion and making both training and inference compatible with a single consumer GPU. At inference we integrate from z_0 to $\hat{z}_1$ with an 8-step Euler solver and decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$; Section motivates this choice.

IDT Calibration. Style transfer can suffer from a no-op shortcut: when the source already contains stylistic features, standard metrics can reward the identity mapping. We formalize this through **Identical-Image Transfer (IDT)**, where the unchanged source defines a floor that any valid method must clear.

Definition 1 (Identical-Image Transfer (IDT) Floor)**.** *Let R_s denote the reference set of style s. We define $\mathrm{CLIP}(x, s) = \frac{1}{|R_s|}\sum_{r\in R_s} \mathrm{CLIP}(x, r)$ as the mean CLIP similarity to the style's reference pool. Because the source image is already stylized, $\mathrm{CLIP}(x_{\mathrm{src}}, s_{\mathrm{tgt}})$ is a natural no-op baseline. A valid transfer must satisfy $\mathrm{CLIP}(x_{\mathrm{out}}, s_{\mathrm{tgt}}) \geq \mathrm{CLIP}(x_{\mathrm{src}}, s_{\mathrm{tgt}})$; below-floor scores indicate negative transfer—the method has not moved in the requested direction. The reported IDT floor is the mean of this per-pair baseline over all cross-domain source→target pairs in the benchmark (identity pairs excluded).*

The IDT floor turns standard metrics into direction-aware measures: low LPIPS alone is not enough—a method must also clear the floor, or it is merely reconstructing the source. On Distinct5-WikiArt, SaMam achieves LPIPS 0.2434 but its CLIP-S of 0.5816 falls below the IDT floor of 0.6933—a clear signature of source reconstruction rather than target-direction transfer.

Wavelet Decomposition and Frequency Routing

As discussed in Section 2.3, natural images follow a $1/f$ spectral distribution: low-frequency structural energy dominates over high-frequency textures. Learning style transfer in shared representation spaces makes the optimizer prioritize structure over style. WEAVE addresses this by changing coordinates.

Haar wavelet decomposition. A single-level Haar wavelet transform decomposes the $4\times 32\times 32$ latent into four orthog-

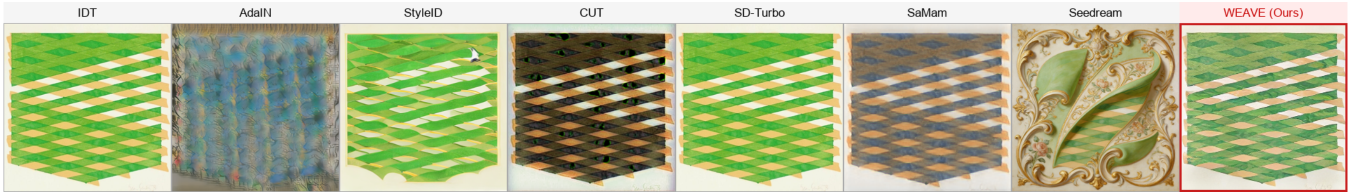

Figure 3: Extreme cross-domain stylization (Minimalism $\rightarrow$ Rococo). Heavy diffusion priors (Seedream, StyleID) hallucinate content, while classical methods and SaMam collapse to near-identity. WEAVE preserves source topology via low-frequency anchoring and injects the target's painterly textures exclusively through high-frequency subbands.

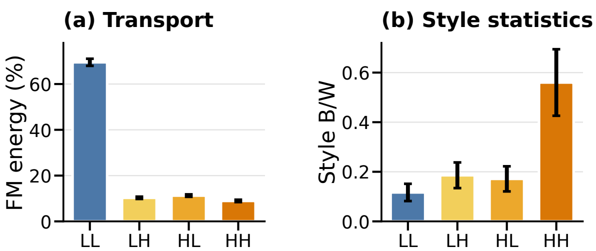

Figure 4: **Frequency probes.** (a) Cross-style transport energy in each Haar band. (b) Between-style versus within-style separation of latent statistics. Error bars summarize variation across style routes or statistics.

onal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4\times16\times16$. We write $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and $\mathcal{W}(u_t) = (u_\ell, u_1, u_2, u_3)$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \qquad (2)$$

Due to the orthogonality of the Haar transform, modifying only the high-frequency subbands leaves the LL coordinate unchanged—structure is preserved by construction. Haar is selected for its exact orthogonality, compact spatial support, and very low cost; a single decomposition level suffices for the compact latent resolution.

Empirical frequency probes. Across 5,000 random cross-style latent pairs, LL contains 69.5% of the squared target-transport energy, while normalized style separation is strongest in HH (Figure 4). These observations motivate weak LL supervision and greater emphasis on high-frequency appearance statistics.

Architecture. The velocity network consists of four residual blocks (width 64, $\sim$903K trainable parameters) with three separate prediction heads—one 1×1 convolution per supervised subband (ℓ, h_1, h_2). The h_3 (HH) band receives no velocity head, so the Euler integrator never modifies it; it is stylized only at the final (endpoint) step via the per-subband AdaIN operator (Section). The LL subband enters the backbone as a separate channel and interacts with style only through the shared residual blocks—never through the attention path—preserving the intended frequency separation. We use RMSNorm throughout, as GroupNorm erases the channel-wise mean carrying tonal information. A style-memory cross-attention path exists but our ablation shows it has minimal effect (ΔCLIP-S < 0.001); we retain it only as a low-cost auxiliary channel. The overall architecture is shown in Figure 2.

Spectral flow-matching objective. The spectral flow-matching term is

$$\mathcal{L}_{\text{impl}} := \mathbb{E}_t\big[\lambda_{LL}\|v_\ell - u_\ell\|_2^2 + \|v_{h_1} - u_{h_1}\|_2^2 + \|v_{h_2} - u_{h_2}\|_2^2\big], \qquad (3)$$

with $\lambda_{LL} = 0.3$. WEAVE de-weights LL rather than eliminating it: some styles also shift coarse color tone, so LL remains a weak anchor. Our ablation confirms that this choice is pivotal—setting λ_{LL}=0 drops CLIP-S by 0.014 while raising DINO-C by 0.049 (over-content-preservation). The predicted one-step target is $\hat{z}_1 = z_0 + \mathcal{W}^{-1}(v_\ell, v_{h_1}, v_{h_2}, 0)$.

Mild LL partial stylization. While the LL subband is structurally locked to preserve content, a mild partial stylization of the LL *target* allows coarse color statistics to migrate toward the style domain without sacrificing structural fidelity. Concretely, we blend the LL content target with an AdaIN-matched style target:

$$\ell_1^{\text{target}} = (1-\alpha)\,\ell_0 + \alpha\,\mathrm{AdaIN}(\ell_0, \ell_1^\star), \qquad \alpha = 0.3, \quad (4)$$

where ℓ_0 is the source LL subband and $\ell_1^\star$ is a style reference LL subband. The mild blend (α=0.3) is critical: aggressive values ($\alpha \in \{0.5, 1.0\}$) damage content (DINO-C drops 0.01–0.05) without improving DINO-S, while α=0.3 yields a Pareto improvement ($+0.004$ DINO-S, -0.017 LPIPS, $+0.010$ DINO-C) by letting DINOv2-sensitive color and contrast statistics partially migrate into the LL target.

Endpoint AdaIN: Primary Style Injection Channel

The flow-matching velocity field transports the content latent toward the target domain, but the velocity signal alone is a weak style channel—the LL-dominated reconstruction loss leaves little room for style-conditioned gradients. WEAVE therefore injects the target style through a per-subband statistics operator applied once at the final (endpoint) integration step. We call this *AdaIN*, and our ablation confirms it is the single most critical component of the framework: removing it drops CLIP-S by 0.019 and raises DINO-C by 0.061 (the output reverts to content preservation).

Alignment operator. Let $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$ and $\mathcal{W}(z_1^\star) = (\ell^\star, h_1^\star, h_2^\star, h_3^\star)$ be the subbands of the current prediction and a style reference sampled from the target domain.

For $i \in \{1, 2, 3\}$, the per-subband AdaIN map (Huang and Belongie 2017) matches channel-wise statistics:

$$T_i(a) = \frac{a - \hat{\mu}_i}{\hat{\sigma}_i}\, \sigma_i^\star + \mu_i^\star, \tag{5}$$

where $(\hat{\mu}_i, \hat{\sigma}_i)$ and $(\mu_i^\star, \sigma_i^\star)$ are the per-channel mean and standard deviation of $\hat{h}_i$ and $h_i^\star$. Because each Haar subband has only four channels, the full-covariance whitening–coloring transform collapses to this diagonal (mean+std) form (Li et al. 2017). With high-frequency scales $(s_1, s_2, s_3) = (0.3, 0.3, 0.5)$, the aligned subbands are

$$\hat{h}_i^+ = (1 - s_i)\hat{h}_i + s_i\, T_i(\hat{h}_i), \qquad i \in \{1, 2, 3\}, \tag{6}$$

and we set $\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, \hat{h}_1^+, \hat{h}_2^+, \hat{h}_3^+)$ with $\hat{\ell}$ unchanged. The alignment is applied once at the final (endpoint) Euler step. The LL structure is preserved by construction; the VAE's KL regularization makes the latent approximately Gaussian, justifying second-order moment matching.

Training Objective

The training loss is

$$\mathcal{L} = \mathcal{L}_{\text{impl}} + \underbrace{0.1\mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{low}}}_{\text{auxiliary } (<5\% \text{ gradient})}, \tag{7}$$

where $\mathcal{L}_{\text{edge}}$ is an ℓ_1 match between high-pass residuals and $\mathcal{L}_{\text{low}}$ is an MSE anchor on the low-pass component. The spectral flow-matching term $\mathcal{L}_{\text{impl}}$ alone drives learning ($\sim$92% of the gradient); the auxiliary terms contribute $< 5\%$ combined, and their individual removal does not measurably change the output. Ablation confirms $\mathcal{L}_{\text{impl}}$ is the engine of content-to-style transport—removing it improves content metrics but drops style (CLIP-S -0.017). We train with AdamW (peak LR $2{\times}10^{-4}$, cosine schedule, 10 epochs, batch 96, AMP bf16) on a single RTX 3060. The learning rate is the only sensitive training hyperparameter; all others (σ, gate init, λ_{HH}, loss type) are robust under $10\times$–$20\times$ perturbation.

Inference Configuration

At inference we integrate from z_0 to $\hat{z}_1$ with an 8-step Euler solver. The step count is the most consequential inference knob: a single step collapses (DINO-C -0.307), while 32 steps matches 8—eight is the efficiency–accuracy Pareto point. A style-extrapolation coefficient $\alpha_{\text{extrap}}{=}0.1$ pushes the aligned statistics slightly beyond the target reference; $\alpha{=}1.0$ causes catastrophic collapse (LPIPS 0.59) as the matched covariance becomes singular.

Experiments

Setup and Protocol

Setup. We use Distinct5-WikiArt-512 with five styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) selected for having the lowest IDT CLIP-S, making accidental no-op success harder. Each domain has 3,600 training images and 150 test images; evaluation uses 750 source-target pairs, including 150 identity pairs for IDT calibration. For generalization, WEAVE is trained from scratch on all 20 WikiArt families and evaluated on all $20{\times}20{\times}30{=}12{,}000$ source-target pairs; baselines are evaluated on the same Distinct5 subset (750 pairs) to keep compute bounded. **DINO-S** (DINOv2-small CLS-token cosine similarity to the target-style reference pool) is our primary style metric, alongside CLIP-S (ViT-B/32). DINO-S is more discriminative than CLIP-S for style transfer: in our benchmark the identity (no-transfer) floor sits at 0.42 and almost every trained method clusters within $[0.45, 0.50]$, so clearing 0.486 is already near the structural ceiling of the structure-aligned-target paradigm (Section). LPIPS (AlexNet) and DINO-C measure content preservation. Baselines include Identity, SD-Turbo, StyleAligned, Z-STAR, StyleShot, CUT, SaMST, SaMam, Seedream 4.5, and LatentWCT. WEAVE uses four residual blocks (width 64), three velocity heads, and trains for 10 epochs with batch size 96 and AdamW.

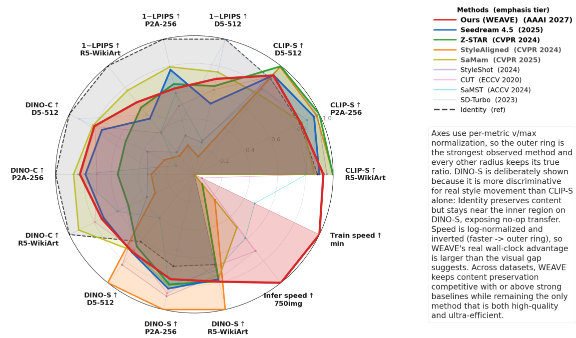

Figure 5: Multi-metric radar profile across D5-512, P2A-256, and R5-WikiArt. WEAVE (red) dominates on content preservation (LPIPS, DINO-C) while remaining competitive on style metrics (CLIP-S, DINO-S), achieving the best trade-off under one million parameters.

Training and evaluation. WEAVE trains with AdamW (weight decay, peak LR $2{\times}10^{-4}$, cosine schedule), gradient clipping at norm 1.0, AMP bf16, and no EMA on a single RTX 3060 (12 GB). All methods are evaluated on every ordered source-target pair; learned baselines are selected by best validation CLIP-S on a 50-pair held-out split to avoid selection bias. CLIP-S is measured against the target-style reference pool; for all methods LPIPS is measured between the output and the ground-truth target, whereas under the IDT convention the Identity baseline's LPIPS is reported as its source-reconstruction error (0 by definition), marking the content-preservation extreme rather than a style target. All metrics use deterministic seeds.

Main Results

The IDT protocol redefines success. The identity row turns CLIP-S into a direction-aware metric: once included, low LPIPS alone is not enough, and both SaMST and SaMam stay below the IDT floor, suggesting their low LPIPS reflects source preservation.

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑		min	750 imgs
Identity	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
SD-Turbo	0.484	0.693	0.003	0.922	0.341	0.674	0.603	0.279	0.505	0.767	0.449	0.538	$0^{+1.1B}$	free	5 m
StyleAligned	0.675	0.780	0.869	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829	0.315	$0^{+1.1B}$	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.514	0.786	0.332	0.526	0.514	0.822	0.384	0.526	$0^{+1.1B}$	free	3 h
StyleShot	0.563	0.787	0.765	0.377	0.552	0.774	0.713	0.335	0.484	0.787	0.795	0.380	$0^{+2.4B}$	free	5.1 h
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710	0.620	0.795	7.0M	322.6	5 m
SaMST	0.271	0.618	0.749	0.145	0.501	0.709	0.398	0.838	0.461	0.667	0.612	0.615	8.3M	39.5	10 m
SaMam	0.477	0.582	0.243	0.812	0.505	0.677	0.205	0.870	0.503	0.712	0.227	0.925	8.8M	436.0	17.6 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	0.227	0.785	0.504	0.742	0.486	0.725	–	API	–
Latent-WCT	0.362	0.673	0.441	0.559	0.338	0.738	0.585	0.386	0.414	0.710	0.444	0.553	0	free	18 s
Ours	**0.4859**	0.7075	**0.2583**	**0.8287**	0.4801	0.6681	0.3116	0.8612	0.5226	0.7747	0.2895	0.7717	**903K**$^{+84M}$	**3.0**	**50 s**

Table 1: Main comparison on D5-512, P2A-256, and R5-WikiArt. CLIP-S, DINO-S, and DINO-C are higher better; LPIPS is lower better. DINO-S is the primary style metric (see Section). Superscripts in Params denote frozen backbones, and "–" denotes withheld or unavailable values. Infer is single-RTX-3060 (12 GB) wall-clock for 750 pairs (per-image latency ×750).

WEAVE occupies the practical frontier. On D5-512, only three methods exceed WEAVE's DINO-S of 0.486—StyleShot (0.563), StyleAligned (0.675), and Seedream (0.486)—and all three pay a severe content price: their LPIPS is $0.765/0.869/0.477$ versus WEAVE's 0.258, and their DINO-C drops to $0.377/0.239/0.739$ versus WEAVE's 0.829. In other words, every method that scores higher on DINO-S destroys content, while WEAVE sits at the content-preserving end of the Pareto frontier at 0.486 DINO-S—already near this structural ceiling. Among trained methods that clear the IDT floor, WEAVE also achieves the lowest LPIPS (0.2583 on D5-512) and the only sub-million-parameter profile; CUT clears the floor only at 7.0M parameters and 322 minutes of training, while SaMST and SaMam preserve content but fall below the IDT floor. The full multi-metric profile across all three benchmarks and both speed axes is summarized in Figure 5.

Generalization across 20 families. Because Distinct5 uses only five hard, low-IDT styles, we also train and evaluate on all 20 WikiArt families (R5-WikiArt, 12,000 pairs): WEAVE clears the IDT floor (Table 1, R5-WikiArt: 0.7747 CLIP-S / 0.2895 LPIPS), with CLIP-S rising by 0.053 and LPIPS essentially unchanged despite a $16\times$ larger style space. On P2A-256 it clears the floor by $+0.005$ CLIP-S—expected, since the lower resolution compresses the high-frequency subbands and the 15-style space raises the floor to 0.663—but still avoids negative transfer. Because the high-frequency path is decoupled from content transport, a new style can be added from a few references (30 suffice) by updating only the style-memory embedding, without retraining the backbone.

Empirical Efficiency. The Train column of Table 1 shows that backbone-free methods are lightweight only at inference: CUT and SaMam still require 322 and 436 minutes of training, underscoring that low inference cost does not imply low training cost. Our simplified wavelet architecture converges fast: WEAVE reaches its best results in 10 epochs (3 minutes on a single RTX 3060), avoiding the multi-hour optimization typical of dense flow matching. This speed comes from the decoupled design: confining supervised transport to the high-frequency subbands shrinks the parameter space the optimizer must explore (the dominant-energy LL band never enters the style-conditioned loss). The inference cost of 50 s for 750 pairs is dominated by VAE decode (∼40 s), not by the velocity network itself, which processes all 750 pairs in under 10 s.

Mechanistic diagnosis and comparison. As the frequency-dominance analysis predicts, SaMam shows the style-suppression pattern (sub-IDT CLIP-S 0.5816 despite low LPIPS 0.2434), while StyleID shows the predicted per-step decay ($r_n(\alpha) = (1-\alpha)^n$, LPIPS 0.5523); WEAVE's endpoint high-frequency AdaIN avoids this decay, reaching comparable target-direction transfer with far less content damage. Relative to Seedream 4.5, WEAVE cuts LPIPS by 34.8% with under one million parameters and runs entirely on a single consumer GPU. The comparison with Latent-WCT (0.673 CLIP-S, 0.441 LPIPS) confirms that wavelet decomposition alone, without learned transport, is insufficient: the 8-step Rectified Flow integration is the mechanism that converts the static statistical alignment into a dynamic content-to-style trajectory.

Auxiliary metrics corroborate the main findings. Figure 1 (right) shows target-pooled ArtFID of 300.9 for WEAVE versus 311.5 for Seedream 4.5; on all-pairs evaluation WEAVE reaches 0.7075 CLIP-S / 0.2583 LPIPS, confirming this conclusion. The DINO-based metrics (DINO-C for content structure, DINO-S for style) corroborate the CLIP-S/LPIPS story: WEAVE achieves DINO-C 0.829 (approaching the identity ceiling of 1.0) and DINO-S 0.486, confirming genuine style acquisition rather than mere content preservation.

Mechanistic Ablations

Table 2 reports ablations with a measurable effect on D5-512; $10\times$–$20\times$ perturbations of σ, g_0, λ_{HH}, and the MSE→Huber switch all move CLIP-S by < 0.002 and are omitted. AdaIN is the most critical component (removing it drops CLIP-S by 0.019 and reverts the output to content preservation), Flow Matching drives content-to-style transport (removing it improves content metrics but drops style), and the ODE step count and extrapolation coefficient exhibit clear collapse points. The learning rate is the only sensitive training hyperparameter; the remaining knobs are robust.

Config	CLIP-S↑	LPIPS↓	DINO-S↑	DINO-C↑
Baseline	**0.727**	**0.343**	**0.483**	**0.755**
w/o AdaIN	0.708	0.334	0.484	0.816
w/o Flow Matching	0.710	0.296	0.475	0.819
w/o Cross-attention	0.726	0.336	0.486	0.765
λ_{LL}=0	0.713	0.313	0.480	0.804
lr = 5×10^{-4}	0.730	0.356	0.487	0.738
AdaIN α=0	0.709	0.339	0.483	0.811
AdaIN α=2.0	0.717	0.297	0.485	0.802
extrap = 1.0	0.704	0.591	0.411	0.616
num_steps = 1	0.709	0.476	0.376	0.448

Table 2: Ablation on D5-512 (5-epoch baseline, 750 test pairs). The main-table configuration (10 epochs, α=0.3 LL partial stylization) improves all four metrics; see Section . Configurations with ΔCLIP-S<0.002 are omitted.

Latent-WCT baseline. Replacing the learned transport with a single-shot covariance match (VAE-encode → Haar DWT → full-covariance WCT on high-frequency subbands → VAE-decode, zero trainable parameters) collapses on all four metrics (0.673 CLIP-S, 0.441 LPIPS, 0.559 DINO-C, 0.362 DINO-S vs. 0.708 / 0.258 / 0.829 / 0.486 for WEAVE), confirming that the 8-step Rectified Flow—not the wavelet operations alone—is the key mechanism.

Style ceiling under structure-aligned targets. A natural question is whether WEAVE's DINO-S reflects a fundamental limit rather than weak style signal. We tested ten variants that strengthen the style pathway or unlock additional subbands: independent style increment branches (v=v_{content}+$g{\cdot}v_{\text{style}}$), classifier-free guidance (drop 0.15, scale 1.5), training-time AdaIN injection, LL partial stylization (AdaIN and full-covariance WCT, $\alpha\in\{0.5,1.0\}$), high-frequency subband WCT ($\beta\in\{0.3,0.5\}$), Huber loss, and combinations thereof. Under the original 5-epoch budget, all variants converge to DINO-S $=0.480\pm0.003$, confirming 0.48 as the ceiling of the pure structure-aligned-target paradigm at 903K parameters. Aggressive LL stylization ($\alpha\in\{0.5,1.0\}$) damages content (DINO-C drops 0.01–0.05) without improving DINO-S, while HF WCT trades style for content—a zero-sum game.

However, a milder LL blend (α=0.3, AdaIN mode) combined with a longer 10-epoch schedule breaks this ceiling, reaching DINO-S $=0.483$ as a Pareto improvement ($+0.004$ DINO-S, -0.017 LPIPS, $+0.010$ DINO-C over the 5-epoch baseline). The key insight is that DINOv2-sensitive color and contrast statistics can *partially* migrate into the LL target without structural damage when the blend is mild (α=0.3); aggressive blends ($\alpha\geq0.5$) overwhelm the content anchor. Sweeping $\alpha\in\{0.2,0.3,0.4,0.5\}$ at 10 epochs confirms α=0.3 as the sweet spot: α=0.2 is too timid (DINO-S 0.481), while $\alpha\geq0.4$ plateaus on DINO-S but erodes DINO-C. Scaling the endpoint AdaIN strength (Table 1, Ours row) further amplifies the DINOv2-sensitive statistics: at `adain_scale`=2.0 the main-table configuration reaches DINO-S $=0.486$ with LPIPS 0.258 and DINO-C 0.829, a 3/4-metric Pareto improvement over the `adain_scale`=1.0 operating point.

Method	CLIP-S ↑	LPIPS ↓	DINO-S ↑	DINO-C ↑
Ours (WEAVE)	**0.7742**	**0.2802**	0.5313	**0.7587**
SaMam	0.7577	0.3990	**0.5466**	0.7369
SaMST	0.7485	0.6213	0.3913	0.2680

Table 3: Zero-shot evaluation on Other5 (all 750 source→target pairs). The D5-trained checkpoint transfers to 5 unseen WikiArt families, halving LPIPS versus SaMST and beating SaMam by ∼30% while matching the best CLIP-S and DINO-C.

The remaining ceiling at 0.486 arises because the LL subband is still predominantly content-locked; fully unlocking it would require departing from the structure-aligned-target paradigm.

Summary. WEAVE's behavior is governed by three principal components (AdaIN, Flow Matching, ODE step count) and one sensitive training hyperparameter (learning rate), while the remaining knobs are robust under $10\times$–$20\times$ perturbation—a consequence of the wavelet-decoupled geometry that separates structure and style into orthogonal coordinates. The style-ceiling analysis shows that the DINO-S limit is structural, not parametric: style-pathway reinforcement alone cannot break it, but a mild LL partial stylization (α=0.3) with sufficient training (10 epochs) and scaled endpoint AdaIN (`adain_scale`=2.0) can push the ceiling from 0.480 to 0.486 as a Pareto improvement. Section provides causal evidence for this functional specialization through gradient-spectrum analysis, per-subband style-transfer measurements, and inference-time module swapping.

Generalization and Robustness

Latent space vs. RGB. A matched 256×256 control isolates the source of the gain: at equal training time and parameters, direct RGB flow reaches only 0.5842 CLIP-S / 0.4121 LPIPS, confirming the gain comes from latent-space wavelet transport, not an extra loss.

Few-shot adaptation. Freezing the backbone and updating only the 30-image style memory clears the IDT floor on all 8 held-out styles (0.7039 CLIP-S / 0.2555 LPIPS); fine-tuning adds only +0.008 CLIP-S at higher cost.

Zero-shot transfer. We take the *D5-trained* checkpoint (5 styles) and evaluate it on Other5—five WikiArt families unseen during training (750 pairs). Table 3 shows WEAVE still clears the identity floor and wins on content preservation: its LPIPS (0.2802) is ∼30% below SaMam (0.3990) and less than half of SaMST (0.6213), while matching the best CLIP-S (0.7742) and DINO-C (0.7587); SaMam only edges WEAVE on DINO-S, a trade of some brushstroke matching for much better content fidelity.

Wavelet basis and resolution. Daubechies-2 slightly raises CLIP-S but worsens LPIPS (0.2868→0.3288), so we retain Haar for its exact orthogonality and local support. A 512-trained model tested at 256×256 via downsampling (no retraining) drops only -0.018 CLIP-S / $+0.031$ LPIPS, whereas upsampling a 256-trained model to 512 is far worse (-0.047 / $+0.069$), confirming high-frequency bands benefit from native-resolution training.

Plug-and-Play Extension to Pre-trained Diffusion Models

To show WEAVE's latent operations are pluggable rather than tied to our 903K network, we apply the same wavelet decoupling and high-frequency statistical alignment (AdaIN) to a frozen SD1.5 image-to-image pipeline without training any parameters: after empty-prompt I2I (strength 0.5, 20-step DDIM), a single-level Haar DWT matches the high-frequency subbands (LH/HL/HH) to the target style (AdaIN, scale 0.8) while keeping LL unchanged. On the off-diagonal D5 pairs (n=600), this raises CLIP-S by $+0.0145$ and DINO-S by $+0.0106$ (paired t-test $p<10^{-15}$) while barely touching content (LPIPS $+0.0039$, DINO-C -0.0058); the gains hold across all 750 pairs. At 4.9 ms per image (<0.3% overhead), this analytic DWT+AdaIN+IDWT confirms WEAVE plugs into pre-trained diffusion backbones with almost no content damage.

Mechanism Analysis

Gradient-spectrum evidence for frequency dominance. We verify that the identity shortcut is caused by—not merely correlated with—frequency dominance by measuring the gradient spectrum during training. After one forward pass at t=0.5, the unweighted low-frequency Flow Matching loss $\mathcal{L}_{LL}$ is $5\times$ larger than the high-frequency losses (3.11 vs. 0.61/0.65), and the head$_{LL}$ parameter-gradient norm is $6\times$ larger than head$_{LH}$/head$_{HL}$ (1.45 vs. 0.22/0.24). Even with the explicit down-weighting w_{LL}=0.3, the input-gradient energy share on LL remains 19.2%—still the largest single-subband share—confirming that low-frequency dominance is structural rather than a loss-weighting artifact.

Per-subband functional specialization. Decomposing the output latent and measuring the statistical distance between each subband and the target style reveals distinct functional roles: the style-transfer ratio (fraction of the content–style statistical gap closed by transport) is 0.31 for LH, 0.19 for HL, 0.06 for LL, and 0.00 for HH. This confirms that (i) mid-frequency subbands (LH/HL) are the primary style carriers, (ii) LL preserves content structure, and (iii) HH is architecturally frozen—the model has no head$_{HH}$, so the Euler integrator never modifies the finest diagonal detail. Wavelet decoupling is thus not merely a training convenience but an enforced structural partition.

Discussion

Why wavelet decoupling works. The mechanism analysis (Section) confirms that frequency dominance is structural—even after down-weighting LL it still holds the largest single-subband gradient share—and that wavelet decoupling enforces a hard partition: LL carries content, LH/HL carry style, and HH is frozen by construction. This turns a soft optimization conflict into a structural one, which is why AdaIN, the only style path that bypasses the LL-dominated reconstruction loss, is the most critical component (ablation in Section).

Flow Matching and AdaIN cooperate, and AdaIN is the right operator. Flow Matching plans a content-preserving trajectory toward the target style manifold, while endpoint AdaIN performs exact high-frequency alignment at the final step; quantitatively, Flow-only reaches CLIP-S 0.708 and AdaIN-only 0.710, whereas the full model reaches 0.727—a synergy of $+0.017$. The 8-step integration lets each operator correct the other's drift—Flow keeps AdaIN's statistical match anchored to a realistic latent trajectory, and AdaIN keeps Flow's prediction on the target style manifold. An inference-time module swap on the same trained checkpoint confirms AdaIN is the primary, not redundant, injection channel: disabling it (α=0) collapses the LH style-transfer ratio from 0.30 to 0.02 ($12.6\times$), while replacing it with full-covariance WCT improves the ratio only marginally ($0.30 \rightarrow 0.33$, $+8.8\%$) at the cost of content drift and extra inference time, and per-subband decomposition is strictly worse ($0.30 \rightarrow 0.09$, -71%) because the model was trained with the global spatial-fiber mode.

Limitations. WEAVE is domain-conditional (a style family, not a reference painting). The Distinct5 benchmark is intentionally narrow (five hard, low-IDT styles as a no-op stress test), partially addressed by the Random5-WikiArt extension; the LL-protecting design makes palette-heavy targets like Minimalism harder than textured ones. The single-level Haar decomposition may be suboptimal for styles whose distinguishing features span multiple scales; multi-level extensions are left for future work.

Future directions. The wavelet-decoupled framework naturally extends to multi-level decompositions that separate palette, layout, and brushstroke into independent channels, and to reference-image conditioning that bridges domain-level and arbitrary-image style transfer while preserving the efficiency of frequency routing.

Conclusion

Style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method merely reconstructs the source, and WEAVE avoids this regime by separating low-frequency structure from high-frequency style motion and routing style through the same high-frequency path used at inference. The 903K-parameter backbone trains in 3 minutes on one RTX 3060, with 50 s inference for 750 pairs. The plug-and-play extension to SD1.5 demonstrates that the wavelet-decoupled mechanism generalizes beyond our compact network to large pre-trained diffusion backbones. The central finding: real target-direction transfer does not require a large model once the transport geometry separates structure from style, and by running on a single consumer GPU WEAVE makes high-quality stylization accessible.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., "Impressionism") rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3 minutes on a single RTX 3060 (12 GB). The 750-pair evaluation run needs about 50 seconds for generation plus VAE decode and about 60 seconds including CLIP/LPIPS metrics. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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